

ALEJANDRO HERNANDEZ

POWER SYSTEMS ENGINEER & DATA SCIENTIST

Power Systems Engineer and Data Scientist with 8+ years connecting renewable energy projects and large inverter-based loads (IBL) to European grids across Spain, the UK, and Denmark. I lead and deliver the full spectrum of grid connection studies. I work at the interface between engineering and other stakeholders such as TSOs, DSOs, developers, and consultants, translating technical requirements into design inputs. With a PhD in machine learning for smart distribution grids and an MSc in Data Science, I pair power system expertise with strong software skills, developing Python tools, analytical frameworks that make engineering insight accessible to decision-makers.
I am looking for roles on Grid Connection, leading and developing grid studies teams that unites advanced power system studies, customer interaction, and product development.

CONTACT

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https://ahhm14.github.io

EDUCATION

2020 - 2025
P.H.D ELECTRICAL ENGINEERING
Polytechnic University of Catalonia

2019 - 2020
MASTER OF FUNDAMENTALS OF DATA SCIENCE
University of Barcelona

2013 - 2018
MECHANICAL ELECTRICAL ENGINEERING
Tecnologico de Monterrey

SKILLS

- Power System Analysis: DIgSILENT PowerFactory, PSCAD, PSS®E
- Python: Automation of power system studies, data processing and analytics
- Machine Learning & AI: Scikit-learn, TensorFlow, Foundation Models
- Engineering Design: AutoCAD
- Project Delivery: Communication, project management, scientific reporting

LANGUAGES

- Spanish (Native)
- English (Professional)
- German (Basic)

WORK EXPERIENCE

Ramboll - Barcelona, Spain — 06/ 2025 - PRESENT
Senior Power System Specialist
- Lead power systems engineering studies for renewable projects with focus on offshore wind farm grid connections across Europe.
- Led and developed Python-based analysis tools and PowerFactory automation libraries to accelerate grid code studies.
- Managed grid studies for large inverter-based loads and project deliverables, within multi-disciplinary projects.
- Prepared technical proposals for a wide variety of consulting and engineering services within the power systems scope.

Power System Specialist — 10/ 2023 - 06/2025
- Conducted steady-state, dynamic stability, and time-series analysis in compliance with several European TSOs.
- Performed techno-economic feasibility studies for early-stage renewable, storage, and data center projects.
- Conducted technical and economic analysis for HVDC interconnection concept design, including converter station placement and cable routing optimization in compliance with ENTSO-E TYNDP.
- Led and authored technical papers for CIGRE conferences and events.

CITCEA - UPC - Barcelona, Spain — 07/ 2020 - 09/2023
Project Engineer/ PhD Candidate
- Conducted machine learning research for power system optimization within European research projects, focusing on distribution grid congestion management.
- Employed ML frameworks for congestion forecasting and demand-side flexibility optimization, published findings in high-impact Journals.
- Analyzed the technical impact of local energy communities in smart distribution grids, with focus on congestion and flexibility services.
- 3 month research stay and collaboration at NTNU and SINTEF Energy, Trondheim Norway.

DIRAM SA de CV - Monterrey, Mexico — 05/2017 - 06/2019
Project Engineer
- Development of electrical analysis studies such as power flow, short circuit, and harmonics for industrial plants in Mexico.
- Data analysis, simulation, and grid compliance studies for solar and wind utility-scale power plants (up to 400 MW). Automatized grid simulations for the assessment of interconnection requirements.
- Design and on-site installation of reactive power compensation equipment such as capacitor banks and harmonic filters.
- Field experience in data collection on-site and power quality and electrical measurements in high, medium, and low voltage levels.
- Management and support for EPC clients during interconnection projects to transmission grid for large-scale industrial and generator plants.